

DON DE CASTRO

Senior Technical Artist / Motion Capture

Los Angeles, CA | dondecastro@gmail.com | [LinkedIn](#) | [Portfolio](#)

Senior technical artist with 20 years across film, games, and virtual production. Equally at home in real-time tools, on a motion capture stage, and inside a character simulation workflow. Spent the last eight years on the Avatar franchise running motion edit, designing retargeting workflows, and partnering with engineering on novel capture problems. Currently animating an independent short in Unreal Engine 5 and writing small C++ editor tools using Claude Code.

Focus: Real-time animation and motion capture work where DCC, engine, and pipeline meet, with a track record of building the workflows teams actually use under deadline pressure.

CORE COMPETENCIES

Real-Time / Unreal Engine: UE5, Sequencer, Blueprint, Control Rig, Animation Blueprints, editor plugin development · **Pipeline & Tools:** Python, Shell, MEL, LLM-native plugin authoring (Claude Code), workflow automation · **Motion Capture & Virtual Production:** On-set supervision, retargeting, Simulcam, ingestion and delivery pipelines, MotionBuilder · **Character FX:** Cloth, hair, and creature simulation (Qualoth, Marvelous Designer, Katana); parallel discipline across career

PROFESSIONAL EXPERIENCE

Technical Director / Developer 2025 to Present

Independent Short Animation Project (Unreleased) | Los Angeles, CA

- Setting up character rigs for the short's hero characters in Unreal's Control Rig, including adapting existing Maya rigs into the Control Rig system
- Animating directly in Sequencer, including shot assembly and camera work
- Wrote a couple of small C++ editor tools using Claude Code that globally override keyboard shortcuts to speed up the animation workflow

Motion Edit Supervisor Sept 2017 to Aug 2025

Lightstorm Entertainment, Avatar: The Way of Water & Avatar: Fire and Ash | Manhattan Beach, CA

- Directed on-set motion capture operations for two feature films across eight years of continuous shooting and post-production
- Designed a retargeting workflow for significant actor-to-CG character scale differences, including a dynamic solution that tracked a growing child actor across years of production
- Identified the technical approach for underwater performance capture, working with developers to solve novel data challenges (splashes, bubbles, refractions, surface-crossing artifacts) with no existing industry reference
- Designed the ingestion and delivery workflows serving internal teams and external vendors across multi-year production
- Oversaw full-load motion stitching and cleanup, combining performance segments into continuous loads for maximum creative flexibility
- Collaborated daily with Production, Live Action, Sequence Leads, and Assets teams to troubleshoot and hit delivery milestones

WDI Visual Ride Designer Jan 2016 to Sept 2017

Lightstorm Entertainment, Pandora: The World of Avatar (Walt Disney World) | Manhattan Beach, CA

- Created animated creature vignettes for projection mapping onto physical props, bridging digital and practical elements for Disney's Flight of Passage attraction
- Partnered with the Production Designer and Ride Design Supervisor on narrative and technical integration

Motion Capture Consultant Nov 2014 to Sept 2016

Sony PlayStation | Playa Vista, CA

- Provided hands-on pipeline training to Sony's MoCap crew on Lightstorm's motion capture software and workflows; workflows from that training carried through to Marvel's Spider-Man and God of War in their shipped form
- Freelance motion editing across external studio projects

Motion Edit Supervisor Jan 2010 to Nov 2014

Giant Studios | Los Angeles, CA

- Designed the Virtual Production Pipeline and Simulcam workflow used across multiple AAA game and film projects
- Wrote Shell and MEL automation that removed manual workflow steps and accelerated delivery
- Managed complex rig assimilation: adapting external rig formats into proprietary software, then delivering refined motion back to the source format
- Supervised final character motion quality and approval across all production phases; primary technical liaison on set

Motion Edit Supervisor Apr 2006 to Oct 2009

Giant Studios, Avatar (Original) | Los Angeles, CA

- Led the team of editors that generated all Na'vi character motion for the original Avatar
- Defined the ingest and delivery pipeline transferring motion data to Lightstorm's Virtual Lab; the workflow shape continued in use for years afterward

Character FX (Concurrent Discipline) 2003 to Present

Cloth, hair, and creature simulation across Sony Imageworks (Hotel Transylvania 2, Alice in Wonderland, Open Season), Blue Sky Studios (Ice Age 5), Blur Studios (game cinematics, commercials, music videos), and Rhythm and Hues (Chronicles of Narnia). Maintained as a parallel discipline alongside motion capture work throughout career.

TECHNICAL SKILLS

Real-Time / DCC: Unreal Engine 5 (Sequencer, Blueprint, Control Rig, Animation Blueprints, Editor Plugins), Maya, 3ds Max, MotionBuilder, Katana

Pipeline / Scripting: Python, Shell Scripting, MEL, Claude Code (LLM-native plugin authoring)

Specializations: Motion capture supervision, virtual production, Simulcam, retargeting, cloth/hair/creature simulation (Qualoth, Marvelous Designer)

EDUCATION

Bachelor of Science, Media Arts and Entertainment

Art Institute of California, Los Angeles, CA | 2003

SELECTED CREDITS

Motion Edit/Capture: Avatar: Fire and Ash, Avatar: The Way of Water, Avatar, Pandora (Disney), Call of Duty: Advanced Warfare, Shadow of Mordor, World of Warcraft, Green Lantern, Real Steel, Chronicles of Narnia, The Polar Express

Character FX: Ice Age 5, Hotel Transylvania 2, Star Wars Jedi: Fallen Order, Rainbow Six Siege, Thor: The Dark World, Halo 2 Anniversary, Elder Scrolls, Dark Souls 2, Alice in Wonderland, Open Season, Chronicles of Narnia